

2 Assignment App Sec DVWA

3/15/2026

107/100 Points

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Anonymous Grading: **no**

Unlimited Attempts Allowed

2/19/2026 to 3/15/2026

✓ **Details**

Assignment: Application Security Testing

Objective

In this assignment, you will:

- Deploy DVWA using Docker
- Exploit multiple vulnerabilities at different security levels
- Analyze defensive mechanisms
- Document your findings professionally in Markdown
- Publish your work on your GitHub profile

This assignment is designed to simulate real-world vulnerability assessment and reporting.

Important Rules

- Perform this lab **only on your local machine**
- Do NOT expose DVWA to the public internet
- Do NOT attack any system other than your own DVWA container
- Unauthorized testing outside this scope is prohibited

Part 1: Install Docker

If Docker is not installed:

Download and install:

<https://www.docker.com/products/docker-desktop/> (<https://www.docker.com/products/docker-desktop/>)

Verify:

```
docker --version
```

Part 2: Deploy DVWA in Docker

Pull Image

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Run Container

```
docker run -d \ --name dvwa \ -p 8080:80 \ vulnerables/web-dvwa
```

Verify

```
docker ps
```

Open in browser:

```
http://localhost:8080
```

Click:

Create / Reset Database

Login:

- Username: admin
- Password: password

Part 3: Vulnerability Testing

Set DVWA Security Level to:

- Low
- Medium
- High

You must Google the attacks of all the Modules and test them against each type / severity (low / medium / high) level of Vulnerability.

Take screenshots of each vulnerability, their exploits, and document them in your report.

Document your report in Github, and keep it in Markdown.

Part 4: Documentation Requirements

For each vulnerability, document:

- Vulnerability Name
- Security Level
- Payload Used

- Result
- Screenshot
- Explanation of why it worked

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You must compare behavior across all three security levels.

Part 5: Docker Inspection Tasks

Run and document the output of:

```
docker ps
```

```
docker inspect dvwa
```

```
docker logs dvwa
```

```
docker exec -it dvwa /bin/bash
```

Inside the container:

```
ls /var/www/html
```

Explain:

- Where application files are stored
- What backend technology DVWA uses
- How Docker isolates the environment

Part 6: Security Analysis Questions

Answer the following:

1. Why does SQL Injection succeed at Low security?
2. What control prevents it at High?
3. Does HTTPS prevent these attacks? Why or why not?
4. What risks exist if this application is deployed publicly?
5. Map each vulnerability to its OWASP Top 10 category.

Part 7: Make the report in Markdown & make it a part of github page.

You would

1. Create a new Github folder

dvwa-security-lab

2. Inside that folder, create a file named:

README.md

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Your Markdown must include:

- Proper headings using #
- Code blocks using triple backticks
- Embedded screenshots
- Structured sections
- Clean formatting

Example Markdown structure:

DVWA Security Lab Report

Environment Setup Description...

SQL Injection

Security Level: Low Payload:

```
1' OR '1'='1
```

Result: ...

Security Level: Medium ...

Analysis ...

4. Push the repository to GitHub.
5. Ensure the repository is public.
6. Add this repository to your GitHub profile pinned repositories.

Part 8: Submission in Canvas

Submit:

1. GitHub repository link along with it's folder

<https://github.com/yourusername/dvwa-security-lab>

Grading Rubric

Criteria	Marks
Docker Deployment	10
Vulnerability Testing (All Modules)	30
Security Level Comparison	10

Criteria	Marks
Docker Inspection Analysis	10
Markdown Quality & Formatting	15
GitHub Repository Setup	10

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Total: 100 Marks

Bonus (+10 Marks)

- Deploy DVWA behind Nginx reverse proxy
- Implement HTTPS using self-signed certificate
- Document difference between HTTP and HTTPS traffic

This assignment evaluates:

- Technical execution
- Security understanding
- Analytical thinking
- Professional documentation skills

You are not just exploiting vulnerabilities.

You are learning how to analyze, report, and communicate security findings like professionals.

<https://github.com/breehaqasim/Cybersecurity-Theory-Tools/tree/main/dvwa-security-lab>  [_ \(https://github.com/breehaqasim/Cybersecurity-Theory-Tools/tree/main/dvwa-security-lab\)](https://github.com/breehaqasim/Cybersecurity-Theory-Tools/tree/main/dvwa-security-lab).



<https://hulms.instructure.com/courses/4979/modules/items/181104>



<https://hulms.instructure.com/courses/4979/modu>

You are unable to submit to this assignment as your enrollment in this course has been concluded.